Muhammad Rumman Aslam

AI/ML Engineer · Software Engineering Student

Pakistan | rummanaslam01@gmail.com | LinkedIn | GitHub | LeetCode | Medium

Education

University of Agriculture, Faisalabad

Bachelor of Science in Software Engineering — Faisalabad, Pakistan | Sept 2024 — Jun 2028

- *Cumulative GPA: 3.69 / 4.00 (Recent Semester: 4.00 / 4.00)*

Industrial Experience

FlyRank : ML Engineering Intern — Remote — Chicago, Illinois, United States | Jun – Sep 2026

[Certificate]

- Processed 78M+ performance records and extracted 9 temporal features using 7-observation rolling/lag windows across clicks, impressions, position, and engagement.
- Trained and compared Logistic Regression and Random Forest classifiers against a rule-based baseline using Precision@50, temporal validation, class balancing, and leakage audits.
- Built an end-to-end content prioritization pipeline that scored and ranked 25K client-content pairs, generating explainable review queues with reason codes and action recommendations for SEO teams.

Research Experience

University of Engineering and Technology Lahore ,Faisalabad Campus

NeuroVision-AI : Multimodal Brain Tumor Diagnosis & Explainable RAG System

- Developed a **PyTorch computer vision pipeline (EfficientNet-B0)** to process the BraTS 2020 dataset across **4 tumor labels**, utilizing exact and perceptual hashing for patient-level dataset splitting to strictly prevent data leakage.
- Evaluated model reliability and diagnostic confidence across all tumor classes, demonstrating that standard validation metrics can overstate locked-test performance and real-world robustness.
- Explored the integration of vision models with large language models (LLMs) for image-conditioned literature retrieval, prioritizing diagnostic safety and critically auditing model outputs for clinical reliability.

University of Agriculture Faisalabad

TrustClinAI: Evaluating the Trustworthiness of Clinical AI

- Designed a **multidimensional model-auditing protocol** to evaluate clinical risk predictors (**XGBoost, Random Forest, Logistic Regression**) across 4 core axes: predictive performance, calibration, explainability, and data robustness.
- Architected a **robustness-testing framework** to quantify prediction stability across 2 simulated degradation scenarios (**missing clinical variables and perturbed measurements**), mitigating catastrophic failure under imperfect real-world conditions.
- Formulated an explainability and failure-analysis pipeline utilizing SHAP values to systematically isolate and interpret clinically costly false-negative predictions.

Selected Projects

Livora AI : A Liver Health Screening Platform — AMD Developer Hackathon Act II

GitHub | Submission

- Engineered a multimodal diagnostic backend (Flask/PostgreSQL) integrating a **Tensorflow based MobileNetV2 CNN**, achieving 97% accuracy in classifying 2D liver ultrasound scans (e.g., HCC, Hemangioma).
- Architected a multi-document OCR extraction pipeline to ingest longitudinal lab reports, mapping temporal biomarkers to deterministic **FIB-4 and APRI clinical calculators** to strictly prevent LLM mathematical hallucinations.
- Deployed a safety-gated generative AI assistant (Fireworks/Groq) configured in strict JSON mode, featuring a deterministic red-flag triage engine that continuously screens queries for acute clinical emergencies.

Robust Multimodal Learning for Neurological Disease Under Missing Modalities (Team Lead)

GitHub | Medium

- Engineered a **PyTorch late-fusion architecture** for cognitive impairment classification across **436** subjects from the OASIS dataset, integrating a **3D CNN** for MRI volumes and a clinical MLP.
- Implemented **random modality dropout** training to simulate real-world data absence, ensuring the network degraded gracefully and maintained predictive stability when imaging or clinical inputs were withheld.
- Audited the pipeline to remove demographic confounds and prevent data leakage, **using 5-fold cross-validation** to rigorously evaluate the fusion model, achieving a stable 73.8% test accuracy.

GraphBRAIN-ABIDE, Graph Neural Networks for ASD Classification (Team Lead)

GitHub

- Led a benchmark comparing **3 classical ML** models against **5 GNN architectures** (GCN-ResNet, GATv2, ChebNet, BrainGNN, Population GCN) modeling functional brain connectomes from **rs-fMRI data.**

- Evaluated **model robustness** against inter-scanner domain shift using site-harmonization techniques and **19-fold Leave-One-Site-Out (LOSO)** cross-validation across 848 subjects and 19 sites from the ABIDE 1 dataset.
- Analyzed out-of-distribution generalization tradeoffs between graph-based representations and traditional feature-based pipelines across multi-site neuroimaging cohorts.

LeafSense AI : Reproducible Deep Learning Plant Agent (Team Lead) Microsoft Agent League Hackathon
GitHub | Live Demo

- Architected a 5-step agentic workflow combining a **MobileNetV2 transfer learning model** (96.7% accuracy across 38 classes) with GPT-4o-mini via GitHub Models API for live, context-aware treatment generation.
- Implemented a deterministic confidence threshold (less than 60% rejection gate) to eliminate out-of-distribution hallucinations and prevent agricultural misinformation.
- Integrated a context-aware sidebar AI agent for post-diagnosis Q&A, automated ReportLab PDF generation, and tracked experiment iterations using Weights & Biases.

The Negotiator : Multi-Agent Autonomous Voice System (Team Lead) HackNation Hackathon
GitHub | Certificate

- Architected a modular full-stack multi-agent platform (Next.js, FastAPI, Supabase) integrating GPT-4o Vision and ElevenLabs conversational agents.
- Designed a deterministic red-flag auditing engine detecting pricing inconsistencies, uncapped hourly rates, and non-itemized quotes.

Skills

Programming Skills: Python, C++, SQL, MySQL, Data Structures & Algorithms.

ML/DL Frameworks Libraries: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy, OpenCV, Matplotlib, Keras, Feature Engineering.

ML/DL Architectures: CNNs, GNNs, Vision Transformers (ViTs), LLMs, Multimodal Fusion, Model Robustness.

Medical AI Computer Vision: Medical Image Analysis (MRI, Ultrasound, Histopathology), Image Segmentation, Explainable AI, Missing-Modality Inference.

Mathematical Foundations: Linear Algebra, Calculus, Probability Statistics, Numerical Optimization, PCA.

Infrastructure MLOps: Docker, Kubernetes, FastAPI, Streamlit, PostgreSQL, Supabase, Weights Biases, MLflow, Git/GitHub, Linux.

Core Competencies: Independent Research, Technical Reporting, Multidisciplinary Teamwork, Failure Analysis.

Volunteering and Teaching

iCodeGuru : Lead Machine Learning Instructor Remote, Silicon Valley, USA | Sept 2026

- **Team Lead:** Leading a 5-member instructional team teaching an intensive, month-long machine learning course to a cohort of 20+ Tech and Non Tech students.
- **Core Curriculum Delivery:** Teach fundamental and advanced concepts in TensorFlow, PyTorch, Scikit-Learn, and deep neural architectures through active code implementation.
- **Practical Work:** Guided students through hands-on projects, code reviews, and systematic debugging sessions to build robust software engineering and ML habits.

Leadership and Achievements

- **Academic Excellence:** Achieved a **4.00/4.00 GPA** in the current semester, ranking **1st across the entire batch**.
- **Scholarships:** Awarded a **fully funded, merit-based bachelor's scholarship** alongside the **Punjab Chief Minister's Merit Laptop**.
- **Team Leadership:** Led an international team with members from **MIT and Munich** during the HackNation Hackathon and also served as a lead instructor in ML course.
- **Instructional Leadership:** Served as lead instructor for an intensive machine learning course, mentoring students and guiding curriculum delivery.
- **Hackathon Recognition:** Delivered outstanding performances in both the **AMD Developer Hackathon** and the **AI Factory Hackathon**.
- **Competitive Programming:** Advanced to the pre-final round of the **International Computer Science Competition (ICSC)** and received a special honor for submitting an exceptionally correct and efficient code solution, and solved **150+ LeetCode Problems**.

Certifications

- **Machine Learning Specialization** — DeepLearning.AI
- **Deep Learning Specialization** — DeepLearning.AI
- **Mathematics for Machine Learning & Data Science** — DeepLearning.AI
- **TensorFlow: Advanced Techniques Specialization** — DeepLearning.AI
- **MLOps Specialization** — Duke University